KHAAD BHARAT: Soil Moisture Preservation Manual

Scientific Study of Biochar Water-Holding Capacities & Carbon Retention

1. The Soil Moisture Challenge

As temperature levels rise across North India during dry spells, irrigation cost increases drastically. Sandy soils (common in Rajasthan and Southern Haryana) suffer from extreme drainage, causing water and urea inputs to leach deep into groundwater, bypassing crop roots. Clay soils compact under heat, choking oxygen from roots.

2. Biochar Water-Holding Mechanics

Biochar contains highly dense micropores (0.5 to 10 micrometers) that trap water through capillary action. This moisture is held against gravity but remains easily extractable by fine crop root hairs. Adding 5% biochar by soil weight creates a permanent carbon sponge that increases overall plant-available moisture capacity by up to 41.5%.

3. Verified Research Results

* **Irrigation Costs:** Agricultural trials showed a 30% to 35% reduction in irrigation frequency, saving diesel pump running costs.
* **Nutrient Retention:** Urea and phosphate retention in sandy root zones improved by 28%, reducing fertilizer runoff.
* **Root Density:** Fine root mass in tomato and chili nurseries increased by 35% within the first 30 days.

4. Summary & Soil Regeneration

Because biochar is highly stable, it does not decompose like compost. It remains active in agricultural soils for decades, making it a highly cost-effective, long-term soil regeneration solution for sustainable farming.